Truncated Noncharacteristic Deformation Lemma

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1 Lemmas from [KS90]

Consider a projective system of abelian groups $(M_n, \rho_{n,p})_n$ indexed by $\mathbf{N}$. One says that this system satisfies the Mittag-Leffler condition (ML for short) if for any $n \in \mathbf{N}$, the decreasing sequence $(\rho_{n,p}(M_p))_p$ of subgroups of M_n is stationary.

For a projective system of abelian groups $(M_n)_n$, we set

$$M_\infty := \varprojlim_n M_n.$$

Consider a projective system of complexes

$$E_n^\bullet \colon \cdots \to M_n^{j-1} \to M_n^j \to M_n^{j+1} \to \cdots , \tag{1}$$

and projective limit

$$E_\infty^\bullet \colon \cdots \to M_\infty^{j-1} \to M_\infty^j \to M_\infty^{j+1} \to \cdots . \tag{2}$$

Denote by

$$\Phi_k \colon H^k(E_\infty^\bullet) \to \varprojlim_n H^k(E_n^\bullet) \tag{3}$$

the natural morphism.

Proposition 1 ([KS90, Proposition 1.12.4]). Assume that for each $j \in \mathbf{Z}$, the system $\left(M_n^j\right)_n$ satisfies the ML condition. Then

(a) for each $k \in \mathbf{Z}$, the map Φ_k is surjective,

(b) if moreover, for a given i the system $\left(H^{i-1}(E_n^\bullet)\right)_n$ satisfies the ML condition, then Φ_i is bijective.

The following is called Kashiwara's lemma.

Proposition 2 ([KS90, Proposition 1.12.6]). Let $(X_s, \rho_{s,t} \colon X_t \to X_s)_{(s,(s \leqq t))}$ be a projective system of sets indexed by $\mathbf{R}$. Assume that for each $s \in \mathbf{R}$, the canonical maps

$$\lambda_s \colon X_s \to \varprojlim_{r<s} X_r$$

and

$$\mu_s \colon \varinjlim_{t>s} X_t \to X_s$$

are both injective (resp. surjective). Then all the maps ρ_{s_0,s_1} $(s_0 \leqq s_1)$ are injective (resp. surjective).

ML procedure for sheaves

Proposition 3 ([KS90, Proposition 2.7.1])**.** Let X be a topological space and let $F \in \mathsf{D}^+(\mathbf{Z}_X)$. Let $(U_n)_{n\in\mathbf{N}}$ be an increasing sequence of open subsets of X, $(Z_n)_{n\in\mathbf{N}}$ a decreasing sequence of closed subsets of X. Set $U \coloneqq \bigcup_n U_n$, $Z \coloneqq \bigcap_n Z_n$.

(i) For any j, the natural map $\varphi_j\colon H^j_Z(U;F) \to \varprojlim_n H^j_{Z_n}(U_n;F)$ is surjective.

(ii) Assume that for a given j, the projective system $(H^{j-1}_{Z_n}(U_n;F))_n$ satisfies the ML condition. Then φ_j is bijective.

(iii) Let now $(X_n)_{n\in\mathbf{N}}$ be an increasing family of subsets of X satisfying $X = \bigcup_n X_n$ and $X_n \subset \operatorname{Int}(X_{n+1})$ for all n. Assume that for a given j, the projective system $\left(H^{j-1}(X_n;F)\right)_n$ satisfies the ML condition. Then the natural map $H^j(X;F) \to \varprojlim_n H^j(X_n;F)$ is bijective.

2 Noncharacteristic Deformation Lemma

Proposition 4. Let X be a locally compact (i.e., by definition, Hausdorff) space countable at infinity (i.e., countable union of compact subspaces). Let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{Z}_X)$, and $(U_t)_{t\in\mathbf{R}}$ be a family of open subsets of X. Let $k \in \mathbf{Z}$. If s is a real number, set $Z_s \coloneqq \bigcap_{t>s} \overline{U_t \setminus U_s}$. We assume the following conditions.

(i) $U_t = \bigcup_{s<t} U_s$, for all $t \in \mathbf{R}$.

(ii) For all pairs (s,t) with $s \leqq t$, the set $\overline{U_t \setminus U_s} \cap \operatorname{supp} F$ is compact.

(iii) For all $s \in \mathbf{R}$ and all $x \in Z_s \setminus U_s$, we have

$$(H^j_{X\setminus U_s}(F))_x = 0 \quad \text{for all } j \leqq k.$$

(iv) For all pairs (s,t) with $s < t$, $Z_s \subset U_t$.

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$H^j\left(\bigcup_{s\in\mathbf{R}} U_s;F\right) \xrightarrow{\sim} H^j\left(U_t;F\right) \quad \text{for all } j<k,$$

and the injections

$$H^k\left(\bigcup_{s\in\mathbf{R}} U_s;F\right) \hookrightarrow H^k\left(U_t;F\right).$$

Proof. Consider the following conditions for $j \in \mathbf{Z}$ and $s \in \mathbf{R}$:

$(a)^j_s$ $\qquad \mu^j_s\colon \varinjlim_{t>s} H^j(U_t;F) \xrightarrow{\sim} H^j(U_s;F)$,

$(a')^k_s$ $\qquad \mu^k_s\colon \varinjlim_{t>s} H^k(U_t;F) \hookrightarrow H^k(U_s;F)$,

$(b)^j_s$ $\qquad \lambda^j_s\colon H^j(U_s;F) \xrightarrow{\sim} \varprojlim_{r<s} H^j(U_r;F)$.

Let us prove, for all $s \in \mathbf{R}$, $(a)_s^j$ for $j < k$ and $(a')_s^k$. We may assume that $\overline{U_t \smallsetminus U_s}$ is compact for any $s \leqq t$ by replacing X by $\operatorname{supp} F$. Consider the distinguished triangle

$$\mathrm{R}\Gamma_{X \smallsetminus U_s}(F) \to F \to \mathrm{R}\Gamma_{U_s}(F) \xrightarrow{+1} .$$

Let us endow with $Z_s \cup U_s$ the induced topology, and denote by $i_s \colon Z_s \cup U_s \hookrightarrow X$ the natural continuous map. Applying the functor

$$(\ \cdot\ )|_{Z_s \cup U_s} = i_s^{-1}$$

to the d.t. above, we have

$$\mathrm{R}\Gamma_{X \smallsetminus U_s}(F)|_{Z_s \cup U_s} \to F|_{Z_s \cup U_s} \to \mathrm{R}\Gamma_{U_s}(F)|_{Z_s \cup U_s} \xrightarrow{+1} .$$

Again applying to this triangle the functor $\mathrm{R}\Gamma(Z_s \cup U_s;\ \cdot\ )$, we have

$$\begin{tikzcd}[column sep=small] & \mathrm{R}\Gamma(Z_s \cup U_s; \mathrm{R}\Gamma_{X \smallsetminus U_s}(F)|_{Z_s \cup U_s}) \arrow[dr] & \\ \mathrm{R}\Gamma(U_s; F) \arrow[ur, "+1"] & & \mathrm{R}\Gamma(Z_s \cup U_s; F|_{Z_s \cup U_s}). \arrow[ll] \end{tikzcd} \tag{4}$$

By hypotheses (iii) and the definition of local cohomology, it follows that

$$H^j_{X \smallsetminus U_s}(F)\Big|_{Z_s \cup U_s} = 0 \quad \text{for any } j \leqq k,$$

and then

$$H^j\left(Z_s \cup U_s; \mathrm{R}\Gamma_{X \smallsetminus U_s}(F)|_{Z_s \cup U_s}\right) = 0 \quad \text{for any } j \leqq k.$$

Thus we have

$$H^j(Z_s \cup U_s; F|_{Z_s \cup U_s}) \xrightarrow{\sim} H^j(U_s; F) \quad \text{for any } j < k, \tag{5}$$

$$H^k(Z_s \cup U_s; F|_{Z_s \cup U_s}) \hookrightarrow H^k(U_s; F). \tag{6}$$

Therefore, if we show Claim 5 below, we have

$$\begin{aligned} \varinjlim_{t>s} H^j(U_t; F) &\cong H^j(Z_s \cup U_s; F|_{Z_s \cup U_s}) \\ &\xrightarrow{\sim} H^j(U_s; F) \end{aligned}$$

for any $j < k$, and $(a)_s^j$ follows. Similarly, we have

$$\begin{aligned} \varinjlim_{t>s} H^k(U_t; F) &\cong H^k(Z_s \cup U_s; F|_{Z_s \cup U_s}) \\ &\hookrightarrow H^k(U_s; F) \end{aligned}$$

and $(a')_s^k$ follows.

Claim 5. For any $j \in \mathbf{Z}$ and $s \in \mathbf{R}$, there exists a natural isomorphism

$$\varinjlim_{t>s} H^j(U_t; F) \cong H^j(Z_s \cup U_s; F|_{Z_s \cup U_s}).$$

Proof of Claim 5 First we prove that for $F \in \mathrm{Mod}(\mathbf{k}_X)$ there exists an isomorphism

$$\varinjlim_{t>s} \Gamma(U_t; F) \cong \Gamma(Z_s \cup U_s; F|_{Z_s \cup U_s}). \tag{7}$$

Consider the following diagram.

$$\begin{tikzcd}
\varinjlim_{t>s} \Gamma(U_t; F) \arrow[d] \arrow[dr] & \\
\varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F) \arrow[r] \arrow[d] & \Gamma(Z_s \cup U_s; F|_{Z_s \cup U_s}) \arrow[d] \\
\varinjlim_{U \supset Z_s} \Gamma(U; F) \arrow[r] & \Gamma(Z_s; F|_{Z_s})
\end{tikzcd}$$

Since $(U_t)_{t>s}$ is a cofinal family of $(U \cup U_s)_{U \supset Z_s}$, we have

$$\varinjlim_{t>s} \Gamma(U_t; F) \xrightarrow{\sim} \varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F),$$

so it is enough to show that

$$\phi \colon \varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F) \to \Gamma(Z_s \cup U_s; F)$$

is isomorphic. Since X is Hausdorff and Z_s is commpact, the natural morhism

$$\psi \colon \varinjlim_{U \supset Z_s} \Gamma(U; F) \to \Gamma(Z_s; F)$$

is isomorphic (see [KS90, Proposition 2.5.1]). Since U_s is open, we have another isomorphism

$$\psi' \colon \varinjlim_{V \supset U_s} \Gamma(V; F) \to \Gamma(U_s; F).$$

The injectivity of ϕ follows from the definition of a sheaf and the injectivities of ψ and ψ'.

Let us prove that ϕ is surjective. Let $\sigma \in \Gamma(Z_s; F)$ be a section of $F|_{Z_s \cup U_s}$. By the isomorphism ψ above, we can extend $\sigma|_{Z_s}$ to an open neighborhood U of Z_s. More precisely, there exists a family $\left(\alpha^U\right)_U \in \varinjlim_{U \supset Z_s} \Gamma(U; F)$ such that

$$\alpha^U\big|_{Z_s} = \sigma|_{Z_s}$$

for all U. Similarly, there exists a family $\left(\beta^V\right)_V \in \varinjlim_{V \supset Z_s} \Gamma(V; F)$ such that

$$\beta^V\big|_{U_s} = \sigma|_{U_s}$$

for all V. These $(\alpha^U)_U$ and $(\beta^V)_V$ satisfy

$$\alpha^U\big|_{U \cap U_s} = \beta^V\big|_{U \cap U_s}.$$

Therefore, by the definition of a sheaf, there exists a unique section $\gamma \in \Gamma(U \cup U_s; F)$ such that

$$\gamma|_U = \alpha^U, \quad \gamma|_{U_s} = \beta^{U_s}.$$

For this γ, we have

$$\gamma|_{Z_s \cup U_s} = \sigma$$

and it follows that

$$\varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F) \to \Gamma(Z_s \cup U_s; F)$$

is an isomorphism.

Next Let $F \in \mathsf{D}^+(\mathbf{k}_X)$. Let $I^\bullet$ be a complex of flabby sheaves quasi-isomorphic to F. Then it follows from (7) that

$$\begin{aligned}
\varinjlim_{t>s} H^j(U_t; F) &\cong \varinjlim_{t>s} H^j\Gamma(U_t; I^\bullet) \\
&\cong H^j\left(\varinjlim_{t>s} \Gamma(U_t; I^\bullet)\right) \\
&\cong H^j\Gamma(Z_s \cup U_s; I^\bullet|_{Z_s \cup U_s}) \\
&\cong H^j(Z_s \cup U_s; F|_{Z_s \cup U_s}).
\end{aligned}$$

This completes the proof. □

Let us come back to the proof of Proposition 4. We prove $(b)^j_s$ for all $j \in \mathbf{Z}$ and $s \in \mathbf{R}$ by induction on j. Since $F \in \mathsf{D}^+(\mathbf{Z}_X)$, we have $(b)^j_s$ for $j \ll 0$ and $s \in \mathbf{R}$. Assume $(b)^j_s$ is proved for all $j < j_0 (< k)$ and $s \in \mathbf{R}$. Since we have $(a)^j_s$ and $(b)^j_s$ for each $j < j_0$ and each $s \in \mathbf{R}$, we can apply Kashiwara's lemma (Proposition 2) and have

$$H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F) \quad \text{for } j < j_0 \text{ and } s \leqq t.$$

Fix $s \in \mathbf{R}$ and consider the projective system $\left(H^{j_0-1}(U_{s-1/n})\right)_{n \in \mathbf{N}}$. This system satisfies ML condition, so, by applying Proposition 3 (iii)[1], we have

$$H^j(U_s; F) \xrightarrow{\sim} \varprojlim_n H^j\left(U_{s-\frac{1}{n}}; F\right).$$

Since $(s - 1/n)_n$ is a cofinal family of $(t)_{s>t}$, we have $(b)^{j_0}_s$. Then by induction on j, we have $(b)^j_s$ for $j < k$ and $s \in \mathbf{R}$. Again by applying Proposition 3 to the projective system $\left(H^j(U_n; F)\right)_n$, we obtain

$$\begin{aligned}
H^j\left(\bigcup_{s \in \mathbf{R}} U_s; F\right) &\cong H^j\left(\bigcup_{n \in \mathbf{N}} U_n; F\right) \\
&\xrightarrow{\sim} \varprojlim_n H^j\left(U_n; F\right)
\end{aligned}$$

[1] [KS90] の証明だと “by Proposition 1.12.4” としているが，Proposition 2.7.1 の誤植かもしれない．どちらでも良いとは思うが，このあと “Again by Proposition 2.7.1” と書いていることからも，誤植だと思う方が自然と思われる．

and for each n, we have by Kashiwara's lemma that

$$H^j\left(U_n; F\right) \cong H^j(U_s; F)$$

for $n \geqq s$.

Finally we replace $(a)_s^j$ by $(a')_s^k$ for $j = k$, and obtain

$$H^k\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \hookrightarrow H^j\left(U_s; F\right).$$

This completes the proof. □

References

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